Image2Traffic: Text-Bridged Cellular Traffic Generation with Reflective Alignment and Temporal Prototypes

Paper #8014

ABSTRACT

Cellular traffic data are essential for network planning and urban computing, yet historical observations are often scarce or unavailable in many regions. Image-conditioned traffic generation addresses this limitation by synthesizing region-specific traffic time series from publicly available satellite imagery and urban contextual information. However, existing methods primarily rely on implicit associations between static visual features and dynamic traffic patterns, leaving cross-modal semantic relationships, fine-grained temporal structures, and diverse traffic patterns insufficiently modeled. In this paper, we propose **Image2Traffic**, a text-bridged and prototype-enhanced diffusion framework for image-conditioned cellular traffic generation. Image2Traffic first employs multi-agent iterative reflection to transform historical traffic series into refined textual descriptions and aligns them with traffic demand semantics inferred from urban imagery through semantic evaluation and refinement. Image2Traffic then constructs frequency-aware temporal prototypes from historical traffic series to complement textual semantics with fine-grained numerical characteristics and periodic patterns. These aligned representations and associated prototypes are organized into a semantic knowledge base, enabling the retrieval and integration of multiple relevant traffic patterns for unseen target regions. Finally, the retrieved semantic and numerical information jointly conditions a diffusion model to generate realistic, region-specific traffic time series. Experiments on three real-world datasets demonstrate that Image2Traffic consistently outperforms competitive baselines, reducing T-JSD by up to 9.97% and daily frequency-component RMSE (DF-RMSE) by up to 8.94%. The code is available at https://anonymous.4open.science/r/image2text/.

CCS CONCEPTS

• **Networks** → **Network simulations**; • **Computing methodologies** → **Modeling and simulation**; • **Information systems** → **Spatial-temporal systems**.

KEYWORDS

Cellular Traffic Generation, Cross-modal Alignment, Prototypes

1 INTRODUCTION

Cellular traffic data capture the spatiotemporal distribution of mobile network usage, providing essential insights into network demand and operational conditions across urban regions [26]. Such data support applications such as network KPI forecasting [23] and resource adaptation in large-scale networks [20]. Related urban data also support region profiling [33] and mobility research [36]. However, obtaining comprehensive cellular traffic data remains challenging because of restricted accessibility, privacy concerns [1], and the absence of historical observations in newly developed or underserved regions. Generative modeling offers a way to synthesize urban mobility data [38] and cellular traffic [4] when observations are limited.

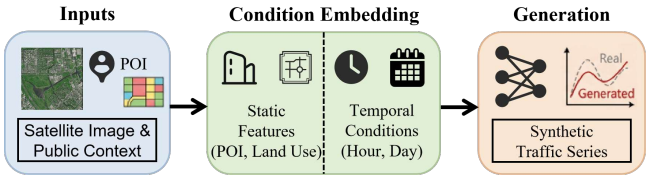


Figure 1: Illustration of image-conditioned cellular traffic generation. Given satellite imagery and publicly available urban contextual information, the task leverages static regional features to synthesize realistic traffic time series.

Existing studies have explored various generative models for cellular traffic, including RNN-based GANs for capturing temporal dependencies [28] and CartaGenie for context-driven traffic generation [30]. In general time-series generation, KoVAE models latent dynamics using Koopman theory [18]. More recently, diffusion-based methods have shown promising performance, such as KSTDiff [40], which uses a knowledge-enhanced diffusion framework for urban flow generation. To reduce reliance on historical traffic observations from target regions, recent studies have shifted toward image-conditioned generation: Open-Diff [4] incorporates satellite imagery, points of interest (POIs), and population information as generation conditions, while DeCAD [31] employs decomposition-enhanced diffusion to synthesize traffic directly from satellite imagery. These advances demonstrate the potential of leveraging publicly available urban imagery and contextual information to generate traffic time series without target-region historical observations. However, satellite imagery primarily captures static urban characteristics, such as building layouts, road networks, and land use, whereas cellular traffic reflects time-varying network demands driven by human activities. Existing image-conditioned methods largely establish these cross-modal relationships through implicitly learned feature associations, leaving the semantic connections between urban environments and fine-grained temporal traffic patterns insufficiently explored.

In this paper, we study image-conditioned cellular traffic generation, aiming to synthesize realistic, region-specific traffic time series from satellite imagery and publicly available urban contextual information without requiring historical traffic observations from target regions, as illustrated in Figure 1. Achieving this objective requires addressing the following three challenges:

• **How to bridge the semantic gap between static urban imagery and numerical traffic time series?** Static urban attributes, including satellite imagery, POI distributions, and population density, describe regional structures and functions, whereas traffic time series record continuous variations in network usage over time. Urban representation learning and mobility prediction integrate heterogeneous regional information [3, 6, 7, 15], but image-conditioned traffic generation additionally requires a connection between static

visual features and dynamic traffic semantics. The first challenge is therefore to establish meaningful semantic alignment between urban visual characteristics and observed traffic time series to guide generation.

• **How to capture fine-grained temporal structures beyond textual representations?** Although textual descriptions can characterize general traffic trends and activity patterns, they often fail to convey precise numerical properties, such as peak intensities, local fluctuations, and temporal periodicities. Language-based approaches have been explored for forecasting, including Time-LLM [11]. Prompt-based forecasting methods such as TEMPO [2] model temporal components, while multi-scale generative models address variations at different resolutions [17]. Preserving these structures in image-conditioned traffic generation requires an explicit numerical representation. The second challenge is therefore to construct temporal representations that complement high-level textual semantics with fine-grained numerical patterns.

• **How to leverage diverse traffic patterns associated with similar urban contexts?** Similar urban environments may exhibit distinct traffic patterns due to variations in human activities, holidays, and special events [35]. Relying on a single representative pattern or coarse-grained conditions may overlook such diversity [9, 21, 29], while directly combining different historical traffic patterns may obscure their distinctive temporal characteristics. The third challenge is therefore to retrieve and integrate complementary temporal information from multiple relevant regions to guide traffic generation.

To overcome these challenges, we propose **Image2Traffic**, a text-bridged and prototype-enhanced diffusion framework for image-conditioned cellular traffic generation. **First, we design a semantic feature alignment module** that employs multi-agent iterative reflection to transform observed traffic series into refined textual descriptions, while an image reasoning agent extracts urban characteristics and infers traffic demand semantics from satellite imagery. By iteratively evaluating and refining their semantic consistency, this module establishes aligned image–text–time-series representations and reusable reasoning templates, thereby bridging the semantic gap between static urban imagery and numerical traffic time series. **Second, we introduce frequency-aware temporal prototypes** that extract representative temporal components from historical traffic series by analyzing their frequency characteristics. These prototypes are associated with the aligned semantic representations obtained in the previous step, complementing textual descriptions with fine-grained numerical patterns and thereby addressing the challenge of preserving temporal structures. **Third, we construct a semantic knowledge base** that organizes the aligned semantic representations and their associated temporal prototypes for similarity-based retrieval. For an unseen target region, the refined reasoning templates first derive traffic demand semantics from its imagery, which are then used to retrieve and aggregate relevant traffic semantics and prototypes from multiple source regions, thereby addressing the one-to-many mapping between urban contexts and traffic patterns. Finally, the integrated semantic and numerical representations jointly condition a diffusion model to generate realistic, region-specific traffic time series.

We summarize our contributions as follows:

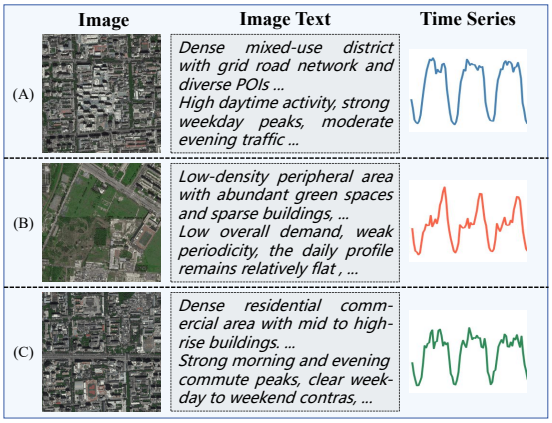


Figure 2: Relationships between satellite imagery and traffic time series. Regions (A) and (B) are distinct urban environments, while regions (A) and (C) are visually similar areas.

• We propose **Image2Traffic**, a text-bridged diffusion framework that establishes semantic alignment between satellite imagery and cellular traffic time series, enabling realistic traffic generation without target-region historical observations.

• We develop a semantic knowledge retrieval mechanism and frequency-aware temporal prototypes to leverage diverse traffic patterns while preserving fine-grained temporal characteristics.

• Extensive experiments on three real-world datasets demonstrate superior generation fidelity and temporal periodicity, particularly in preserving daily and weekly frequency components.

2 PRELIMINARY STUDY

2.1 Image–Text–Traffic Relationships

Urban imagery and cellular traffic provide complementary perspectives on urban regions: satellite imagery captures spatial structures and functional characteristics, while traffic time series reflect temporal variations in network demand. To explore their relationships, Figure 2 presents three representative regions from Beijing, illustrating their satellite images, image-derived semantic descriptions, and observed traffic series. Regions (A) and (B) exhibit distinct urban characteristics, reflected in their textual descriptions of dense mixed-use development and low-density green spaces, respectively, along with substantially different traffic patterns. In contrast, regions (A) and (C) share similar urban semantics and broadly comparable daily traffic patterns, yet differ in their fine-grained temporal variations. These observations suggest that textual descriptions can provide an interpretable semantic connection between urban imagery and traffic patterns, while similar visual semantics do not necessarily imply identical temporal dynamics. This motivates the need to establish cross-modal semantic alignment, capture fine-grained temporal structures, and leverage diverse traffic patterns for image-conditioned generation.

2.2 Problem Definition

We study image-conditioned cellular traffic generation, which aims to synthesize realistic traffic time series for geographic regions using publicly available urban data. Given a region i, we denote its available regional information as $\mathcal{P}_i = \{I_i, O_i, G_i\}$, where I_i, O_i, and

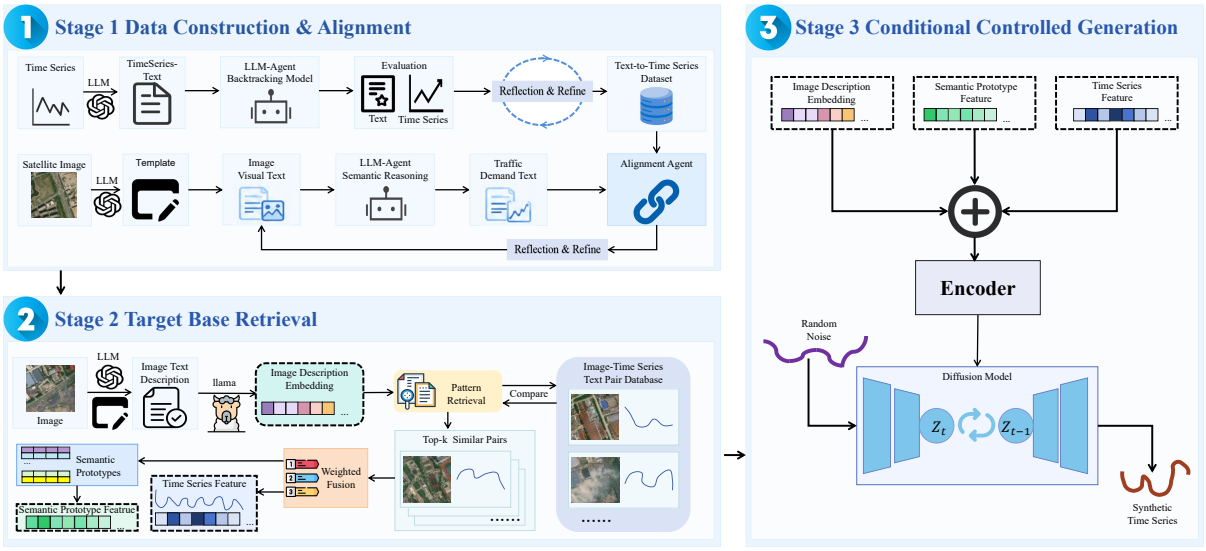


Figure 3: The overall architecture of our framework. It comprises three tightly coupled stages: multimodal semantic alignment, knowledge base retrieval, and conditional diffusion generation. By leveraging retrieved references and semantic prototypes as explicit and implicit conditions, the model achieves high-fidelity and controllable traffic synthesis.

G_i represent satellite imagery, point-of-interest (POI) distributions, and geographic information, respectively. The corresponding cellular traffic is represented as $Y_i = \{y_{i,t}\}_{t=1}^{L}$, where $y_{i,t}$ denotes the aggregated network traffic of all base stations (BSs) within region i at time step t, and L is the sequence length. Formally, the objective is to learn a conditional generative model $p_\theta(Y \mid C)$, where $C_i = f(\mathcal{P}_i)$ denotes the conditioning representation constructed from the available urban data and source-region references. Given paired regional information and traffic observations from source regions for training, the model generates a traffic sequence for an unseen target region q as $\hat{Y}_q \sim p_\theta(Y \mid C_q)$. During inference, only the target region's publicly available information $\mathcal{P}_q$ and the source-region knowledge base are accessible; the target region's historical traffic observations remain unavailable. The generated sequence is expected to capture realistic, region-specific traffic distributions and temporal patterns.

3 METHOD

We design a semantic-alignment and prototype-enhanced diffusion framework for image-conditioned cellular traffic generation. As illustrated in Figure 3, the framework consists of three key components: (1) semantic feature alignment, which converts traffic series and satellite imagery into refined textual descriptions and aligns their semantics; (2) semantic prototype learning, which captures fine-grained temporal structures; and (3) knowledge base retrieval, which retrieves multiple plausible temporal references for each target region. Ultimately, these conditions jointly guide the diffusion generator to synthesize the target traffic series.

3.1 Semantic Feature Alignment

Satellite imagery captures static urban structures, while traffic data reflect dynamic human activities. To bridge this semantic gap, we use text as an intermediate modality, converting both visual and temporal data into aligned textual descriptions.

3.1.1 Time Series Text Generation. We convert traffic time series into textual descriptions through template-based generation and multi-agent refinement. We first construct a time-series (TS) template pool from time-series literature and urban mobility knowledge, removing city- and series-specific details to improve reusability. The templates cover temporal dynamics, periodic patterns, and long-term trends. For each source-region series Y_i, GPT-4o extracts temporal features and fills a sampled template to obtain an initial description $T_{i,0}^{ts}$.

To improve semantic fidelity, we introduce an iterative reconstruction and reflection process involving two agents. We first mask informative segments of Y_i, including peak windows, trough periods, and weekday–weekend transitions. The Temporal Reconstruction Agent reconstructs the masked segments solely from the current text $T_{i,k}^{ts}$, producing $\hat{Y}_i^{\text{mask}}$, and evaluates reconstruction quality using the mean absolute error (MAE) against the ground truth Y_i^{mask}. If the MAE exceeds a threshold τ_{ts}, the TS Reflection Agent identifies missing or inaccurate temporal information and provides targeted revisions to the description and, when necessary, its template. The updated text $T_{i,k+1}^{ts}$ is then evaluated again. This process continues until the MAE falls below τ_{ts} or the maximum number of rounds R is reached. The resulting refined text T_i^{ts} is used for subsequent cross-modal alignment.

3.1.2 Cross-Modal Semantic Alignment. We further align image-derived traffic demand semantics with the refined TS descriptions to bridge static urban characteristics and dynamic traffic patterns. We first construct an image template pool covering observable urban attributes, such as building density, land use, and road structures. For each source-region image I_i, GPT-4o extracts visual features and

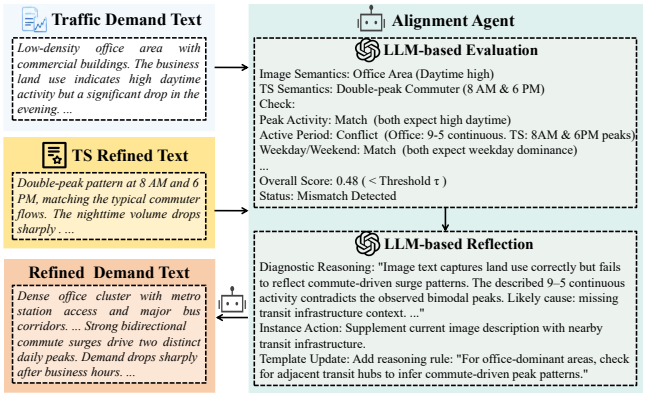


Figure 4: Illustration of the cross-modal semantic alignment and reflection process. The Alignment Agent evaluates the consistency between image-derived descriptions and refined time-series descriptions, while the Image Reflection Agent identifies discrepancies and provides targeted feedback.

fills a sampled template to obtain an initial image description $T_{i,0}^{img}$. Guided by semantic reasoning templates derived from urban planning and mobility literature, an Image Semantic Reasoning Agent then infers traffic demand semantics D_i^{img}, including peak periods, dominant activity purposes, and weekday–weekend differences.

We refine this reasoning process through iterative alignment and reflection. As illustrated in Figure 4, the Alignment Agent compares D_i^{img} with the refined TS text T_i^{ts} across predefined dimensions, such as peak characteristics, active periods, and weekday–weekend differences. An LLM assigns a consistency score to each dimension, and their average yields the overall alignment score a_k. If a_k falls below a threshold τ_{align}, the Image Reflection Agent diagnoses the discrepancies and revises the image description or semantic reasoning template accordingly. The Image Semantic Reasoning Agent then regenerates the traffic demand semantics for the next evaluation round. This process continues until a_k exceeds τ_{align} or the maximum number of rounds is reached, yielding the refined image-derived traffic-semantic text T_i^{img}.

By refining reasoning templates across multiple source regions, the framework aims to correct recurring semantic discrepancies and capture transferable relationships between visual attributes and traffic demand patterns. The resulting template library is reused for unseen regions. During inference, GPT-4o first describes the target image I_q, and the Image Semantic Reasoning Agent applies the refined templates to produce the target image-derived text T_q^{img} without accessing its traffic observations.

3.2 Semantic Prototype Learning

While the retrieved TS texts provide explicit semantic priors for traffic pattern characterization, they describe patterns at a coarse semantic level and cannot precisely encode the periodic structure of time series. Furthermore, different regions sharing similar visual environments may exhibit diverse temporal patterns that text alone cannot fully distinguish. We therefore introduce a semantic prototype representation to capture fine-grained spectral periodicities and temporal structures as an implicit complement to the explicit text-based semantics, enabling more precise and diverse traffic pattern generation.

3.2.1 Semantic Prototype Representation. We define a set of M fixed orthogonal basis vectors $\mathbf{B} = [\mathbf{b}_1, \ldots, \mathbf{b}_M]^\top \in \mathbb{R}^{M \times d}$ as a shared temporal dictionary. The first M_s basis vectors are explicitly designed with semantic meaning. They are derived from normalized temporal templates, including daily and weekly harmonics, high-frequency oscillations, and nocturnal troughs. These templates are projected into $\mathbb{R}^d$ via QR orthogonalization to form the semantic bases. The remaining $M - M_s$ basis vectors span the orthogonal complement, initialized via Gram-Schmidt orthogonalization. All basis vectors are frozen during training to maintain their interpretability. For each source region i during training, we have access to its ground-truth time series Y_i, which is first transformed using a real fast Fourier transform (RFFT). The low-frequency amplitude bins are extracted and normalized as the spectral feature $\mathbf{f}_i \in \mathbb{R}^{C \cdot F}$, where C is the number of traffic channels and F is the number of retained frequency bins covering the daily and weekly harmonics. A lightweight Prototype Assignment Network f_ϕ then maps $\mathbf{f}_i$ to a set of combination weights $\mathbf{w}_i = \text{softmax}(f_\phi(\mathbf{f}_i)) \in \mathbb{R}^M$. Consequently, the prototype embedding for the source region is computed as a weighted combination of the basis vectors, $\mathbf{p}_i = \mathbf{w}_i^\top \mathbf{B} = \sum_{m=1}^{M} w_{i,m} \mathbf{b}_m \in \mathbb{R}^d$.

3.3 Knowledge Base Retrieval

After cross-modal alignment of imagery and time series, we obtain reliable paired representations for each source region, consisting of the refined image-derived text T_i^{img} and the refined TS text T_i^{ts}. However, regions with similar visual environments generally share similar coarse-grained traffic patterns, while still exhibiting diversity in fine-grained temporal behaviors. Therefore, enforcing a strict one-to-one mapping may overlook these local variations. To address this issue, we construct a semantic knowledge retrieval module that enables a one-to-many mapping, allowing a visual environment to retrieve multiple relevant temporal references and capture its fine-grained pattern diversity.

3.3.1 Knowledge Base Construction. The knowledge base stores aligned cross-modal representations from all source regions. For each source region i, a text encoder E_{text} encodes the refined image-derived text and the refined TS text into a visual-semantic embedding and a temporal-semantic embedding, respectively. In addition, the prototype embedding $\mathbf{p}_i$ is computed from the ground-truth series Y_i via spectral feature extraction and the Prototype Assignment Network f_ϕ. Each knowledge base entry is defined as a triplet $\mathcal{K}_i = [\mathbf{e}_i^{img}, \mathbf{e}_i^{ts}, \mathbf{p}_i]$, where $\mathbf{e}_i^{img} = E_{\text{text}}(T_i^{img})$ serves as the retrieval key, and $\mathbf{e}_i^{ts} = E_{\text{text}}(T_i^{ts})$ and $\mathbf{p}_i$ serve as the retrieved values.

3.3.2 Target Reference Retrieval. During inference for a target region q, we extract its semantic query $\mathbf{e}_q^{img}$ from the target refined image-derived text T_q^{img}. We then calculate the cosine similarity between $\mathbf{e}_q^{img}$ and all image embeddings $\mathbf{e}_i^{img}$ stored in the knowledge base. The top-K source regions with the highest similarity scores are retrieved to form a set of temporal-semantic embeddings $\mathcal{E}_q^{ts} = \{\mathbf{e}_{q,1}^{ts}, \ldots, \mathbf{e}_{q,K}^{ts}\}$. To measure the relevance of each retrieved

Table 1: Performance comparison on the Beijing and Shanghai datasets. Baselines are sorted chronologically. Lower values indicate better performance.

Method	T-JSD	T-JSD-FO	S-JSD	S-JSD-FO	CRPS	DF-RMSE	WF-RMSE
Beijing							
TimeGAN	0.1868 ± 0.00482	0.2321 ± 0.00042	0.2669 ± 0.00213	0.1488 ± 0.00120	0.2243 ± 0.00442	0.0483 ± 0.00342	0.0439 ± 0.00085
TimeVQVAE	0.1617 ± 0.00314	0.1907 ± 0.00101	0.2010 ± 0.00243	0.1034 ± 0.00086	0.2112 ± 0.00485	0.0411 ± 0.00287	0.0406 ± 0.00072
ADAPTIVE	0.1439 ± 0.00288	0.1888 ± 0.00058	0.1423 ± 0.00239	0.0973 ± 0.00096	0.2096 ± 0.00467	0.0414 ± 0.00293	0.0403 ± 0.00060
KEGANs	0.1401 ± 0.00254	0.1841 ± 0.00106	0.1384 ± 0.00248	0.0915 ± 0.00025	0.2089 ± 0.00454	0.0407 ± 0.00322	0.0394 ± 0.00097
Open-Diff	0.1315 ± 0.00176	0.1754 ± 0.00054	0.1366 ± 0.00226	0.0921 ± 0.00084	0.1995 ± 0.00384	0.0405 ± 0.00203	0.0396 ± 0.00061
STK-Diff	0.1203 ± 0.00207	0.1653 ± 0.00031	0.1247 ± 0.00204	0.0855 ± 0.00057	0.1957 ± 0.00411	0.0386 ± 0.00295	0.0371 ± 0.00052
BRIDGE	0.1219 ± 0.00184	0.1689 ± 0.00017	0.1203 ± 0.00235	0.0839 ± 0.00028	0.1932 ± 0.00423	0.0394 ± 0.00287	0.0378 ± 0.00075
KGDiff	0.1163 ± 0.00189	0.1644 ± 0.00044	0.1145 ± 0.00198	0.0814 ± 0.00239	0.1864 ± 0.00395	0.0376 ± 0.00143	0.0364 ± 0.00088
Ours	**0.1047 ± 0.00225**	**0.1587 ± 0.00047**	**0.1017 ± 0.00212**	**0.0738 ± 0.00066**	**0.1757 ± 0.00389**	**0.0353 ± 0.00298**	**0.0343 ± 0.00095**
Improvement	9.97%	3.47%	11.18%	9.34%	5.74%	6.12%	5.77%
Shanghai							
TimeGAN	0.2268 ± 0.00421	0.2521 ± 0.00089	0.1800 ± 0.00265	0.1553 ± 0.00041	0.2946 ± 0.00412	0.0642 ± 0.00353	0.0611 ± 0.00102
TimeVQVAE	0.1971 ± 0.00354	0.2353 ± 0.00102	0.1624 ± 0.00214	0.1379 ± 0.00098	0.2793 ± 0.00456	0.0610 ± 0.00332	0.0560 ± 0.00127
ADAPTIVE	0.2039 ± 0.00368	0.2338 ± 0.00085	0.1683 ± 0.00045	0.1323 ± 0.00009	0.2816 ± 0.00478	0.0618 ± 0.00345	0.0596 ± 0.00138
KEGANs	0.1907 ± 0.00312	0.2241 ± 0.00098	0.1634 ± 0.00228	0.1275 ± 0.00024	0.2779 ± 0.00532	0.0595 ± 0.00324	0.0553 ± 0.00115
Open-Diff	0.1803 ± 0.00285	0.2194 ± 0.00167	0.1576 ± 0.00205	0.1221 ± 0.00063	0.2644 ± 0.00495	0.0601 ± 0.00318	0.0587 ± 0.00122
STK-Diff	0.1663 ± 0.00234	0.2053 ± 0.00045	0.1467 ± 0.00189	0.1085 ± 0.00045	0.2467 ± 0.00462	0.0572 ± 0.00205	0.0511 ± 0.00098
BRIDGE	0.1629 ± 0.00215	0.2029 ± 0.00088	0.1453 ± 0.00182	0.1109 ± 0.00052	0.2412 ± 0.00448	0.0591 ± 0.00312	0.0534 ± 0.00104
KGDiff	0.1577 ± 0.00208	0.1969 ± 0.00091	0.1414 ± 0.00175	0.1034 ± 0.00036	0.2385 ± 0.00425	0.0545 ± 0.00396	0.0493 ± 0.00087
Ours	**0.1439 ± 0.00185**	**0.1851 ± 0.00102**	**0.1328 ± 0.00184**	**0.0938 ± 0.00041**	**0.2269 ± 0.00305**	**0.0515 ± 0.00326**	**0.0471 ± 0.00109**
Improvement	8.75%	5.99%	6.08%	9.28%	4.86%	5.50%	4.46%

reference, we compute the normalized retrieval weights $\{w_k\}_{k=1}^K$ based on their similarity scores:

$$w_k = \frac{\exp(\text{sim}(\mathbf{e}_q^{img}, \mathbf{e}_{q,k}^{img}))}{\sum_{j=1}^K \exp(\text{sim}(\mathbf{e}_q^{img}, \mathbf{e}_{q,j}^{img}))}, \tag{1}$$

where $\text{sim}(\cdot,\cdot)$ denotes the cosine similarity. Subsequently, the aggregated temporal-semantic condition for the target region is obtained by a weighted sum of the retrieved embeddings: $\mathbf{e}_q^{ts} = \sum_{k=1}^K w_k \mathbf{e}_{q,k}^{ts}$. This explicit semantic condition $\mathbf{e}_q^{ts}$ is then fused with the implicit numerical prototypes in the subsequent generation stage. Similarly, the prototype embedding $\mathbf{p}_q$ is obtained by aggregating the prototype embeddings of the top-K retrieved references, weighted by their retrieval similarity scores $\{w_k\}_{k=1}^K$: $\mathbf{p}_q = \sum_{k=1}^K w_k \mathbf{p}_{q,k}$, where $\mathbf{p}_{q,k}$ denotes the prototype embedding of the k-th retrieved sample.

3.4 Conditional Diffusion Generation

During training, each source region i provides three aligned conditions: the image-semantic embedding $\mathbf{e}_i^{img}$, the temporal-semantic embedding $\mathbf{e}_i^{ts}$, and the prototype embedding $\mathbf{p}_i$. These conditions are fused to guide the denoising process. The denoising network ϵ_θ and the Prototype Assignment Network f_ϕ are optimized end-to-end by minimizing the standard diffusion loss:

$$\mathcal{L}_{\text{train}} = \mathbb{E}_{Y_i, \epsilon\sim\mathcal{N}(0,\mathbf{I}), t}\left[\left\|\epsilon - \epsilon_\theta\left(Y_i^t, t \mid \mathbf{e}_i^{img}, \mathbf{e}_i^{ts}, \mathbf{p}_i\right)\right\|_2^2\right], \tag{2}$$

where t is the randomly sampled diffusion step and ϵ is the added Gaussian noise. During the inference phase for an unseen target region q, we no longer have access to its ground-truth time series or TS description. Instead, the conditions are constructed from the top-K retrieved references. The target image provides $\mathbf{e}_q^{img}$, while the temporal and prototype conditions are obtained by a weighted sum based on the retrieval similarity scores $\{w_k\}_{k=1}^K$, i.e., $\mathbf{e}_q^{ts} = \sum_{k=1}^K w_k \mathbf{e}_{q,k}^{ts}$ and $\mathbf{p}_q = \sum_{k=1}^K w_k \mathbf{p}_{q,k}$. Starting from pure Gaussian noise, the model iteratively removes noise conditioned on these aggregated features to synthesize the final traffic series $\hat{Y}_q$. The training and inference procedures are summarized in Algorithms 1 and 2, respectively.

4 EXPERIMENTS

We perform evaluations on real-world datasets to answer the following research questions.

- **RQ1:** How does the proposed model compare with existing baselines in terms of generation fidelity?
- **RQ2:** How do image descriptions, time-series descriptions, and prototypes affect generation performance?
- **RQ3:** How effectively does our framework control generation across different urban functional patterns?
- **RQ4:** How do semantic basis vectors affect generation performance?

4.1 Evaluation Settings

4.1.1 Mobile Traffic Datasets. We evaluate our method on three real-world mobile traffic datasets. The Beijing dataset contains traffic records from 4,000 base stations (BSs) over one week in October 2021. The Shanghai dataset covers over 4,000 BSs from August 1 to 7, 2019, while the Nanjing dataset includes over 8,000 BSs from February 2 to 8, 2021. All traffic sequences are sampled at hourly intervals. To align traffic data with satellite imagery, we

Table 2: Impact of image descriptions, time-series descriptions, and prototypes on model performance across three datasets. "w/o" denotes removal of the corresponding component. The best results are highlighted in bold.

Setting	Beijing				Shanghai				Nanjing			
	T-JSD	S-JSD	CRPS	DF-RMSE	T-JSD	S-JSD	CRPS	DF-RMSE	T-JSD	S-JSD	CRPS	DF-RMSE
FULL	**0.1047**	**0.1017**	**0.1757**	**0.0353**	**0.1439**	**0.1328**	**0.2269**	**0.0515**	**0.1482**	**0.1290**	**0.3314**	**0.0601**
w/o img_text	0.1063	0.1119	0.1821	0.0379	0.1581	0.1449	0.2348	0.0602	0.1707	0.1319	0.3438	0.0656
w/o ts_text	0.1123	0.1193	0.1844	0.0368	0.1623	0.1442	0.2339	0.0656	0.1564	0.1391	0.3342	0.0685
w/o proto	0.1077	0.1206	0.1762	0.0356	0.1546	0.1401	0.2367	0.0543	0.1502	0.1297	0.3376	0.0659

Table 3: Performance of conditional mobile traffic generation (T-JSD) for Pattern-0, Pattern-1, and Pattern-2 in each city. Δ denotes the relative reduction compared with KGDiff.

Dataset	Pattern	ResNet	Initial	KGDiff	Ours	Δ
Beijing	Pattern-0	0.1075	0.1062	0.1068	**0.1011**	5.34%
	Pattern-1	0.0926	0.0842	0.0884	**0.0808**	8.60%
	Pattern-2	0.1420	0.1341	0.1248	**0.1157**	7.29%
Shanghai	Pattern-0	0.2165	0.2172	0.2148	**0.2026**	5.68%
	Pattern-1	0.1170	0.1142	0.1153	**0.1094**	5.12%
	Pattern-2	0.2077	0.2041	0.2022	**0.1953**	3.41%
Nanjing	Pattern-0	0.1715	0.1695	0.1709	**0.1622**	5.09%
	Pattern-1	0.1825	0.1842	0.1803	**0.1692**	6.16%
	Pattern-2	0.1384	0.1365	0.1373	**0.1292**	5.90%

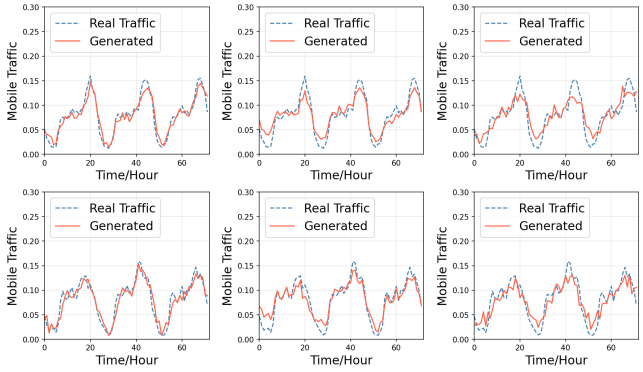
aggregate BS traffic according to the geographic regions covered by the images and apply normalization to the aggregated sequences.

4.1.2 Baselines. We compare our method with eight representative generative baselines. The adversarial baselines are TimeGAN [34], ADAPTIVE [37], and KEGANs [10]. We also include the vector-quantized model TimeVQVAE [14]. The diffusion baselines are Open-Diff [4], STK-Diff [5], BRIDGE [16], and KGDiff [27]. Detailed descriptions are provided in Appendix B.

4.1.3 Metrics. We evaluate generation quality using four categories of metrics: (1) *Distribution fidelity*: T-JSD and S-JSD measure temporal and spatial distribution similarity. (2) *Variation dynamics*: T-JSD-FO and S-JSD-FO assess first-order temporal and spatial variations. (3) *Probabilistic fidelity*: CRPS measures the agreement between generated distributions and real observations. (4) *Periodicity*: DF-RMSE and WF-RMSE evaluate the fidelity of daily and weekly frequency components. Detailed definitions are provided in Appendix C.

4.2 Mobile Traffic Generation (RQ1)

Table 1 reports the generation results on the Beijing and Shanghai datasets, with Nanjing results provided in Table 7. Overall, our model consistently outperforms all baselines across all metrics, demonstrating its effectiveness in generating realistic cellular traffic. In terms of distribution fidelity, our model achieves the lowest T-JSD and S-JSD, improving them by 9.97% and 11.18% over the strongest baseline (KGDiff) on the Beijing dataset, respectively. The lower T-JSD-FO and S-JSD-FO further demonstrate its ability to capture fine-grained spatiotemporal variations, benefiting from cross-modal textual alignment and reference retrieval. In the frequency domain, our model reduces DF-RMSE and WF-RMSE



(a) Ours (b) KGDiff (c) STK-Diff

Figure 5: Visualization of generated samples and generation performance of different models on the Beijing dataset. Our model generates time series with the best fidelity.

by 6.12% and 5.77% on the Beijing dataset, and 5.50% and 4.46% on the Shanghai dataset, respectively. These gains highlight the effectiveness of the frequency-aware temporal prototypes in preserving periodic patterns. Our model also achieves the best CRPS, indicating improved probabilistic fidelity. Figure 5 visualizes the generated traffic sequences on the Beijing dataset. Although existing methods capture overall daily periodicity, KGDiff and STK-Diff exhibit deviations in peak amplitudes and fail to reproduce certain sharp fluctuations. In contrast, our model more closely follows the real traffic curves, preserving both overall temporal trends and fine-grained variations.

4.3 Ablation Study (RQ2)

To evaluate the contribution of each core component, we conduct an ablation study across the three datasets. As shown in Table 2, the full model (FULL) achieves the best performance on all metrics, demonstrating the necessity of all modules. Removing image descriptions (w/o img_text) notably increases both T-JSD and S-JSD across all datasets, indicating that image semantics are crucial for capturing spatial correlations. Removing time-series descriptions (w/o ts_text) leads to clear degradations in CRPS and DF-RMSE, demonstrating the importance of textual temporal constraints. Furthermore, dropping the prototype mechanism (w/o proto) results in higher S-JSD and CRPS, showing the prototype's role in maintaining distributional consistency. In conclusion, all three components synergistically contribute to the high-quality generation.

Table 4: Impact of semantic priors on the Shanghai dataset. "Full" represents the complete set of 12 semantic basis vectors. "Harmonics" and "Non-periodic" represent subsets retaining only periodic harmonics and non-periodic features, respectively. "Random" indicates randomly initialized vectors without semantic information.

Setting	T-JSD	S-JSD	CRPS	DF-RMSE
Full	**0.1439**	**0.1328**	**0.2269**	**0.0515**
Harmonics	0.1452	0.1336	0.2285	0.0519
Non-periodic	0.1453	0.1342	0.2292	0.0524
Random	0.1464	0.1348	0.2306	0.0527

4.4 Conditional Traffic Generation (RQ3)

To evaluate generation controllability across different urban functional zones, we partition the traffic data into three distinct patterns: residential, commercial, and office areas, denoted as Pattern-0, Pattern-1, and Pattern-2, respectively. We extract the corresponding geographical embeddings from satellite imagery to guide the generation. The results (T-JSD) are reported in Table 3. As observed, our model consistently achieves the lowest T-JSD across all three functional patterns and all cities. The ResNet baseline directly extracts spatial features from images without textual descriptions, which limits its ability to represent traffic-related urban semantics. The Initial baseline introduces both image and time-series descriptions but does not apply any optimization to align or fuse these multimodal features. This lack of optimization prevents the model from effectively leveraging the semantic conditions, resulting in suboptimal controllability. KGDiff is competitive but remains less accurate than our method across all evaluated patterns. Our model explicitly optimizes multimodal fusion, enabling it to precisely disentangle fine-grained satellite semantics and generate accurate traffic dynamics tailored to distinct functional zones.

4.5 Analysis of Semantic Prototypes (RQ4)

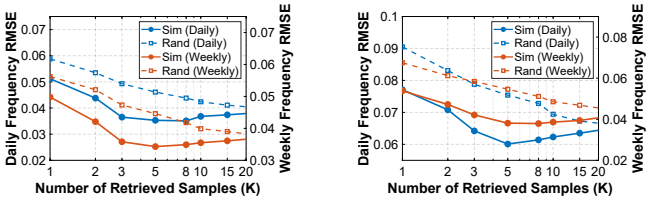
To evaluate the impact of semantic priors, we compare the full set of 12 semantic basis vectors with subsets of 8 harmonic vectors and 4 non-periodic vectors, as well as randomly initialized vectors on the Shanghai dataset. As shown in Table 4, the Full setting achieves the best performance across all evaluation metrics. The Harmonics setting performs comparably, indicating that periodic harmonics are the most critical priors for capturing the fundamental rhythm of traffic generation. However, restricting the model to non-periodic features (Non-periodic) or omitting semantic guidance entirely (Random) leads to performance degradation. This confirms that the structured semantic composition in our full design is highly effective.

4.6 Analysis of the Retrieval Mechanism

To determine the optimal number of retrieved samples K, we conduct a hyperparameter analysis on the retrieval mechanism. In Figure 6, Sim represents our similarity-based retrieval, which selects the top-K historical samples that are semantically closest to the target input, whereas Rand stands for random retrieval, which selects K samples completely at random without any similarity constraints. To mitigate the inherent randomness, the results of Rand are averaged over multiple runs. We compare their performance on DF-RMSE and WF-RMSE across different values of K. As shown in the figure, Sim consistently and significantly outperforms Rand across all evaluated values of K on both metrics and datasets. This indicates that meaningful semantic guidance is crucial, as random selection provides only limited and unstable help. Notably, the error of Sim initially decreases and reaches its lowest point. However, further increasing K leads to a slight performance drop, likely due to the introduction of redundant or less relevant samples. In contrast, Rand shows a gradual improvement as K increases since a larger pool increases the chance of retrieving useful samples, but its performance remains consistently inferior to Sim. Based on the overall performance, we empirically set $K = 5$ for the final model to achieve the best overall generation quality.

Table 5: Cross-city generalization performance. "X-Y" denotes training on X and testing on Y (BJ: Beijing; SH: Shanghai; NJ: Nanjing). $\Delta = (m_{\text{in}} - \max m_{\text{transfer}})/m_{\text{in}}$ measures the worst relative change across transfer settings, where negative values indicate degradation.

Setting	T-JSD	T-JSD-FO	S-JSD	S-JSD-FO	CRPS	DF-RMSE	WF-RMSE
Origin BJ	0.1047	0.1587	0.1017	0.0738	0.1757	0.0353	0.0343
SH-BJ	0.1185	0.1732	0.1086	0.0785	0.1914	0.0419	0.0396
NJ-BJ	0.1203	0.1784	0.1108	0.0801	0.1953	0.0428	0.0413
Δ	-14.90%	-12.41%	-8.95%	-8.54%	-11.16%	-21.25%	-20.41%
Origin SH	0.1439	0.1851	0.1328	0.0938	0.2269	0.0515	0.0471
BJ-SH	0.1642	0.1983	0.1466	0.0993	0.2523	0.0648	0.0517
NJ-SH	0.1615	0.1956	0.1435	0.0947	0.2480	0.0597	0.0429
Δ	-14.11%	-7.13%	-10.39%	-5.86%	-11.19%	-25.83%	-9.77%
Origin NJ	0.1482	0.2498	0.1290	0.0835	0.3314	0.0601	0.0381
BJ-NJ	0.1576	0.2672	0.1410	0.0878	0.3591	0.0714	0.0432
SH-NJ	0.1567	0.2677	0.1389	0.0906	0.3481	0.0669	0.0407
Δ	-6.34%	-7.17%	-9.30%	-8.50%	-8.36%	-18.80%	-13.39%



(a) Beijing **(b) Nanjing**

Figure 6: Impact of retrieval size K on DF-RMSE and WF-RMSE across different datasets. Sim denotes similarity-based retrieval, and Rand denotes random retrieval.

4.7 Robustness

To verify the robustness of our model, we evaluate two cross-city transfer settings for each target city among Beijing (BJ), Shanghai (SH), and Nanjing (NJ). Table 5 presents the results. It can be seen that the performance gaps across different transfer settings are not substantial, indicating that the model maintains stable generation capabilities under domain shifts. Although transferring models to unseen cities may cause certain performance drops in specific metrics, the overall degradation remains within an acceptable range.

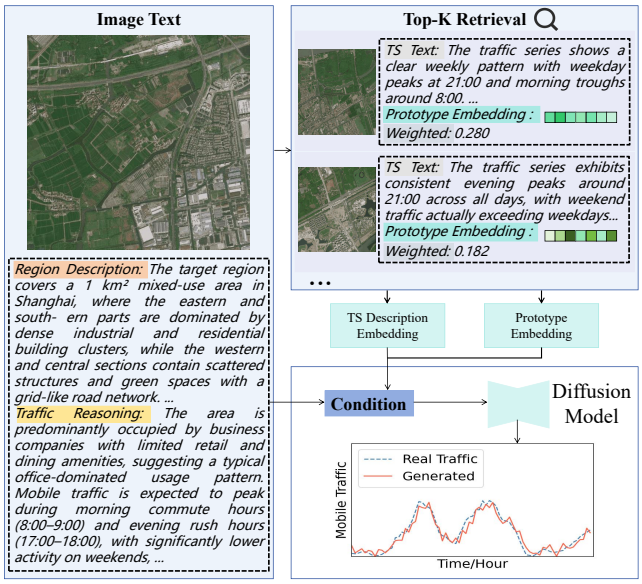


Figure 7: Case study of the generation process, illustrating image-based semantic inference, reference retrieval, and traffic synthesis.

Overall, this demonstrates the strong robustness and generalizability of the algorithm. This robustness is primarily attributed to the prototype mechanism and multimodal alignment. By effectively aligning satellite imagery with textual descriptions, our model captures universal geographical and traffic correlations rather than overfitting to city-specific patterns.

4.8 Case Study

To intuitively demonstrate how our framework synthesizes traffic for an unseen region, we present a case study in Figure 7. Given only the satellite image and POI profile of the target region in Shanghai, our framework generates its traffic time series. During the generation phase, we first generate the region description and infer traffic patterns. Based on this, the knowledge base retrieves the top-K references by comparing image-derived semantic embeddings. Their TS text embeddings and prototype embeddings are aggregated via weighted fusion to form the final condition vector. Guided by these conditions, the diffusion model generates a traffic series that aligns closely with the real data.

4.9 Efficiency Analysis

We further evaluate the deployment efficiency, as summarized in Table 6. We record memory usage, training time, and generation time for a batch of 32 samples. Our model significantly reduces memory usage and training time compared to KGDiff, owing to its lightweight prototype assignment and efficient condition integration. Although the generation time is slightly longer due to the iterative denoising process, this trade-off is acceptable for achieving high-quality synthesis. Overall, our model offers a practical balance between generation fidelity and deployment efficiency.

Table 6: Deployment efficiency comparison.

Model	Deep Learning Framework	Memory Usage (MB)	Training Time (s)	Generation Time (s)
KGDiff	PyTorch 2.4	3898	0.496	0.781
Ours	PyTorch 2.4	2250	0.443	0.869

5 RELATED WORK

5.1 Using Text for Time Series Modeling

Textual modalities have been widely adopted to augment traditional time series modeling, particularly in tasks such as classification [12, 24] and forecasting [32]. These paradigms typically employ natural language to provide semantic context or auxiliary guidance for interpreting fixed sequential data. Meanwhile, synthesizing time series directly from textual descriptions remains an emerging area. One line of research focuses on leveraging LLMs for synthesis. For instance, SDForger [22] maps series to text to fine-tune LLMs and decode new series, while GenG [39] utilizes LLM-generated captions to condition generative models for domain-specific synthesis. Another line of work emphasizes latent space alignment, such as T2S [8], which combines a length-adaptive VAE with text-conditioned latent diffusion to generate time series from prompts. These methods address general text-conditioned synthesis; our framework specifically connects satellite-derived spatial semantics with fine-grained cellular traffic dynamics.

5.2 Conditional Time Series Generation

Conditional diffusion models have emerged as a powerful paradigm for time series generation. Recent works have explored various conditioning directions to enable flexible synthesis. For instance, in data imputation, models are conditioned on partial observations to reconstruct missing values [25]. Another direction focuses on constrained generation, where models are guided to synthesize series that satisfy specific statistical or physical constraints [19]. Beyond time series, TabDDPM [13] models heterogeneous numerical and categorical features for tabular data generation. In contrast, our work introduces a diffusion framework that conditions on both textual descriptions and satellite imagery to enable controllable spatiotemporal generation.

6 CONCLUSION

In this paper, we investigate image-conditioned cellular traffic generation, aiming to synthesize realistic traffic sequences from satellite imagery and POI information without requiring historical traffic data from target regions. To bridge the semantic gap between static urban environments and dynamic traffic patterns, we propose Image2Traffic, a diffusion-based framework that aligns image and traffic semantics through textual representations. Building on this alignment, we incorporate semantic reference retrieval to leverage traffic patterns from similar regions and frequency-aware prototypes to explicitly capture temporal periodicities. Extensive experiments on three real-world datasets demonstrate that our framework consistently outperforms existing methods in distribution fidelity, spatiotemporal dynamics, and periodicity preservation. Future work will explore finer spatial and temporal resolutions and improved generalization across diverse urban environments.

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SUPPLEMENTARY MATERIAL

A TIME SERIES GENERATION ALGORITHMS

The training and inference procedures are summarized in Algorithms 1 and 2, respectively. Here, T is the number of diffusion steps, $\alpha_t = 1 - \beta_t$, $\bar{\alpha}_t = \prod_{u=1}^{t} \alpha_u$, and σ_t is the sampling noise scale.

Algorithm 1: Training the conditional diffusion model.

Input : Distribution of original data $q(Y^0)$
Diffusion steps T
Noise schedule $\{\beta_t\}_{t=1}^{T}$
Conditioning information $\mathbf{e}_i^{img}, \mathbf{e}_i^{ts}, \mathbf{p}_i$
Output : Trained denoising network ϵ_θ and prototype assignment network f_ϕ

while ϵ_θ *has not converged* **do**
  for *source region* i *in dataset* $\mathcal{D}$ **do**
    $Y_i^0 \sim q(Y_i^0)$;
    $t \sim \text{Uniform}(\{1, \ldots, T\})$;
    $\epsilon \sim \mathcal{N}(0, \mathbf{I})$;
    Extract conditions: $\mathbf{e}_i^{img}, \mathbf{e}_i^{ts}$ and compute
    $\mathbf{p}_i = \text{softmax}(f_\phi(\mathbf{f}_i))^\top \mathbf{B}$;
    Update θ and ϕ using the gradient:
    $\nabla_{\theta,\phi} \left[\left\| \epsilon - \epsilon_\theta(Y_i^t, t \mid \mathbf{e}_i^{img}, \mathbf{e}_i^{ts}, \mathbf{p}_i) \right\|^2 \right]$;

Algorithm 2: Inference for a target region.

Input : Noise $Y_q^T \sim \mathcal{N}(0, \mathbf{I})$
Target region q, trained denoising network ϵ_θ
Output : Generated traffic series $\hat{Y}_q = Y_q^0$

for $t = T, T-1, \ldots, 1$ **do**
  if $t > 1$ **then**
    Sample $z \sim \mathcal{N}(0, \mathbf{I})$;
  else
    $z = 0$;
  Obtain conditions: $\mathbf{e}_q^{img}, \mathbf{e}_q^{ts}, \mathbf{p}_q$ for region q;
  $Y_q^{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(Y_q^t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(Y_q^t, t \mid \mathbf{e}_q^{img}, \mathbf{e}_q^{ts}, \mathbf{p}_q) \right) + \sigma_t z$;

B BASELINE MODELS

TimeGAN. The baseline [34] is a classic GAN-based framework that combines adversarial training with a supervised loss and an embedding network to capture the temporal correlations of time series.

TimeVQVAE. The baseline [14] is a VQ-VAE-based generative model that utilizes vector quantization and a bidirectional prior to generate high-quality time series by capturing both low-frequency and high-frequency components.

ADAPTIVE. The baseline [37] is a deep transfer learning framework that leverages an Urban Knowledge Graph (UKG) and a base-station alignment scheme to transfer knowledge across cities for cellular traffic generation.

KEGANs. The Knowledge-Enhanced GAN (KEGANs) [10] is a hierarchical GAN framework that utilizes a self-constructed Urban Knowledge Graph (UKG). The UKG embeddings are concatenated with noise vectors so that the GAN implicitly captures the relationship between conditional information and network traffic.

Open-Diff. The baseline [4] is a diffusion-based model that utilizes publicly available satellite imagery to generate mobile traffic. It adopts a multi-positive contrastive learning module to extract regional features and an attention mechanism to fuse conditions.

STK-Diff. The baseline [5] is a spatiotemporal knowledge-driven diffusion model. It constructs a UKG to characterize urban features and designs a temporal extraction module and a spatial connection module to capture the correlations between mobile traffic and urban environments.

BRIDGE. The baseline [16] is a text-controlled time-series generation framework. It introduces a multi-agent optimization pipeline to synthesize text-TS pairs and integrates semantic prototypes with textual descriptions for domain-level guidance.

KGDiff. The baseline [27] uses a knowledge graph to guide heterogeneous spatiotemporal generation. It encodes urban structural knowledge with a geometric mixture-of-experts and normalizes node-specific heterogeneity.

C EVALUATION METRICS

T-JSD. The Temporal Jensen-Shannon Divergence (T-JSD) measures the distribution similarity of the traffic time series within each region. The T-JSD can be formulated as:

$$\mathcal{J}_{\mathrm{T}} = \sqrt{\frac{KL(P_r(\hat{s}^i) \| P_r(s^i)) + KL(P_r(s^i) \| P_r(\hat{s}^i))}{2}} \tag{3}$$

where P_r denotes an empirical probability distribution and KL denotes the Kullback–Leibler divergence; s^i and $\hat{s}^i$ denote the real and generated mobile traffic sequences of region i, respectively (the Y_i and $\hat{Y}_i$ used in the main text).

T-JSD-FO. This metric compares the distributions of first-order temporal differences. For a sequence of length L, the differences are $z^i = \{s_j^i - s_{j-1}^i\}_{j=2}^{L}$, with $\hat{z}^i$ defined analogously for generated data. T-JSD-FO is computed from $P_r(\hat{z}^i)$ and $P_r(z^i)$ using the same distribution-comparison procedure as T-JSD.

S-JSD. This metric compares the overall traffic distributions across regions. Let B be the number of regions and L the sequence length. We flatten the real and generated traffic matrices of shape $B \times L$ into vectors S and $\hat{S}$, respectively, and compare their empirical distributions $P_r(S)$ and $P_r(\hat{S})$.

S-JSD-FO. Following the same first-order differencing procedure, we obtain Z and $\hat{Z}$ from the flattened real and generated traffic vectors, respectively. S-JSD-FO compares the distributions $P_r(Z)$ and $P_r(\hat{Z})$.

CRPS. The continuous ranked probability score (CRPS) measures agreement between a predictive cumulative distribution and the corresponding observations. Lower values indicate better probabilistic fidelity.

DF-RMSE. We evaluate daily periodicity by comparing the relative magnitudes of daily frequency components. For a generated traffic series $\hat{S}$ of length L sampled at hourly intervals, we compute

Table 7: Performance comparison on the Nanjing dataset. Baselines are sorted chronologically.

	Nanjing						
Method	**T-JSD**	**T-JSD-FO**	**S-JSD**	**S-JSD-FO**	**CRPS**	**DF-RMSE**	**WF-RMSE**
TimeGAN	0.2236 ± 0.00485	0.3241 ± 0.00012	0.1853 ± 0.00364	0.1249 ± 0.00085	0.4218 ± 0.00645	0.0780 ± 0.00285	0.0505 ± 0.00042
TimeVQVAE	0.1884 ± 0.00452	0.2903 ± 0.00025	0.1634 ± 0.00212	0.1088 ± 0.00058	0.3786 ± 0.00724	0.0716 ± 0.00335	0.0467 ± 0.00115
ADAPTIVE	0.1897 ± 0.00415	0.2924 ± 0.00097	0.1622 ± 0.00201	0.1096 ± 0.00024	0.3805 ± 0.00695	0.0724 ± 0.00328	0.0466 ± 0.00108
KEGANs	0.1851 ± 0.00388	0.2885 ± 0.00102	0.1587 ± 0.00254	0.1081 ± 0.00095	0.3774 ± 0.00615	0.0709 ± 0.00315	0.0458 ± 0.00095
Open-Diff	0.1769 ± 0.00449	0.2801 ± 0.00081	0.1514 ± 0.00239	0.1045 ± 0.00072	0.3612 ± 0.00425	0.0715 ± 0.00385	0.0446 ± 0.00075
STK-Diff	0.1644 ± 0.00365	0.2675 ± 0.00095	0.1396 ± 0.00254	0.0983 ± 0.00042	0.3465 ± 0.00488	0.0688 ± 0.00152	0.0425 ± 0.00038
BRIDGE 2	0.1628 ± 0.00258	0.2687 ± 0.00082	0.1384 ± 0.00148	0.0952 ± 0.00035	0.3487 ± 0.00475	0.0704 ± 0.00248	0.0432 ± 0.00055
KGDiff	0.1601 ± 0.00312	0.2654 ± 0.00057	0.1356 ± 0.00415	0.0926 ± 0.00018	0.3432 ± 0.00385	0.0660 ± 0.00232	0.0405 ± 0.00046
Ours	**0.1482 ± 0.00285**	**0.2498 ± 0.00066**	**0.1290 ± 0.00292**	**0.0835 ± 0.00078**	**0.3314 ± 0.00342**	**0.0601 ± 0.00221**	**0.0381 ± 0.00038**
Improvement	7.43%	5.88%	4.87%	9.83%	3.44%	8.94%	5.93%

$\hat{F} = \mathrm{FFT}(\hat{S})$. The relative daily component is:

$$\hat{F}^d = \frac{\|\hat{F}[k_d]\|_2}{\|\hat{F}\|_2}, \qquad k_d = \frac{L}{24}. \tag{4}$$

Here, k_d is the daily frequency bin and $\|\hat{F}\|_2$ is the ℓ_2 norm of the spectrum. We compute F^d analogously for the real series. The per-series error is $\sqrt{(\hat{F}^d - F^d)^2}$, and DF-RMSE summarizes the root-mean-square error over the evaluated series.

WF-RMSE. We evaluate weekly periodicity in the same way, using the weekly frequency bin $k_w = L/(24 \times 7)$. The relative weekly component is:

$$\hat{F}^w = \frac{\|\hat{F}[k_w]\|_2}{\|\hat{F}\|_2}. \tag{5}$$

We compute F^w for the real series and summarize the errors $\hat{F}^w - F^w$ using RMSE.

D GENERATION ON THE NANJING DATASET

We also evaluate the generation performance on the Nanjing dataset, as shown in Table 7. Consistent with the main experiments, our model achieves the best performance across all metrics. This further demonstrates the strong generalizability of our framework.

E AGENT PROMPT TEMPLATES

The prompt templates used in this work are detailed as follows.

Image Description Prompt

Template: This 1 km × 1 km satellite image of [CITY] depicts a [LAND_USE_TYPE] area. [SPATIAL_DESCRIPTION of building density and distribution]. The road network [ROAD_DESCRIPTION]. The area contains [POI_COUNT] points of interest, with [TOP_CATEGORY_1], [TOP_CATEGORY_2] , and [TOP_CATEGORY_3] as the dominant categories. [ADDITIONAL_FACILITIES] further characterize the functional diversity of this urban environment.
Sample: This 1 km × 1 km satellite image of Shanghai depicts a mixed-use area. Dense industrial and residential building clusters dominate the eastern and southern parts, while scattered structures and green spaces occupy the western and central sections. The road network forms a grid-like layout. The area contains 170 POIs, with companies, shopping, and transportation facilities as the dominant categories.

Figure 8: Prompt for initial image description.

Time Seires Description Prompt

Template: This base station traffic series exhibits a [PERIODICITY_STRENGTH] [PERIODICITY_TYPE] pattern. Peak traffic occurs during [PEAK_TIMING] (around [PEAK_HOUR]) with [PEAK_INTENSITY] intensity, while the lowest activity is observed around [TROUGH_HOUR]. [WEEKDAY to WEEKEND CONTRAST SENTENCE]. The series has a [BASELINE_LEVEL] baseline level and a [TREND_DIRECTION] overall trend.
Sample: This base station traffic series exhibits a clear weekly periodicity pattern. Peak traffic occurs during evening hours (around 21:00) with high intensity, while the lowest activity is observed around 8:00 in the morning. Weekday traffic is notably higher than weekends (ratio=1.37), suggesting a predominantly work-related usage pattern. The series has a moderate baseline level and a stable overall trend.

Figure 9: Prompt for source origin time series description.

Traffic Reasoning Prompt

Template: The area is classified as a [ZONE_TYPE], given [POI_EVIDENCE]. This [FUNCTIONAL_DESCRIPTION] indicates a [PERIODICITY_STRENGTH] daily periodicity driven by [ACTIVITY_DRIVER]. The peak period occurs during [PEAK_TIME], [PEAK_REASON]. Overnight traffic is [BASELINE_LEVEL], and weekend traffic is [WEEKEND_PATTERN] compared to weekdays. The overall volatility of traffic is expected to be [VOLATILITY_LEVEL], reflecting [VOLATILITY_REASON].
Sample: Classified as a business-dominated zone, given high-density company POIs (81) and limited retail/dining. This office environment indicates strong daily periodicity driven by regular commuters. Peak traffic occurs during commute hours (8:00–9:00, 17:00–18:00). Overnight traffic is very low, and weekend activity is significantly reduced compared to weekdays. Overall volatility is expected to be moderate, reflecting predictable patterns associated with office-area commuting behavior.

Figure 10: Prompt for traffic reasoning to initial image text.

Semantic Alignment Prompt

Template: You are a geospatial semantic analyst. Given the following information about an urban region, assess whether its visual characteristics are consistent with its traffic behavior.
Region description: [REGION_TEXT_DESC]
Traffic pattern: [TRAFFIC_TEXT_DESC]
Task: Determine whether the spatial features described are semantically aligned with the observed traffic dynamics. Output a brief alignment judgment and explanation.
Output format:
Alignment: [Consistent / Partially consistent / Inconsistent]
Reason: [1-2 sentences explaining the alignment or discrepancy]
Reflection: [1-2 sentences re-examining the judgment, considering if spatial features are underweighted, traffic patterns have alternative explanations, or confidence should be adjusted]

Figure 11: Prompt for semantic alignment.